OrchScope on the 2011 Sipscan: Testing the Orchestration Engine on a Documented Botnet Campaign

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Abstract

OrchScope's orchestration engine decides whether a campaign of co-tooled scanners is orchestrated, that is, more synchronized than independent operation explains. The paper validates it on simulated campaigns and on random groups of real sources; neither is a real orchestrated campaign. This report tests it on one: the sipscan, a scan of the entire IPv4 space for SIP servers that the Sality botnet ran under central command in February 2011, as captured by the UCSD Network Telescope and released by CAIDA. We replay the scan through ORION-sized views of that telescope, with the paper's test and decision threshold and with peers borrowed from ORION's port-5060 sources. The engine flags the botnet at the first or second hourly update after its onset and flags its restart at the first decision its weekly window allows, but not the 5.7-day steady phase in between, whose timing matches independent scanning. Random groups of ORION's scanners are flagged 5 times in 1,600. The replay also exposed a flaw of the engine's first version, which drew population-null peers from residue sources too: on ORION's port 5060, where 89% of the sources form one residue event, it could flag nothing. OrchScope now excludes residue sources from its peer pools; admitting them would turn the paper's 906 orchestrated campaigns into 884, and both of its negative controls flag none either way.

1 Introduction

OrchScope [1] infers scanning campaigns at the Merit ORION network telescope [2] and decides, for each campaign, whether its sources are orchestrated: whether they act together more tightly than independent operation explains. Its orchestration engine (Section III-C of the paper) compares a campaign's mean pairwise co-activity with two nulls and declares the campaign orchestrated only if both reject. The paper evaluates the engine on synthetic campaigns injected into real background traffic, where the schedule of every bot is known, and on random groups of real sources, which should not be flagged. Both are needed, but neither is a real orchestrated campaign observed in the wild.

This report supplies one. In February 2011, the Sality botnet scanned the entire IPv4 address space for SIP servers (UDP port 5060) under the control of its command-and-control (C&C) infrastructure. The scan, known as the sipscan, reached the UCSD Network Telescope from about three million bots; Dainotti et al. analyze it in detail [3, 4], and CAIDA releases every sipscan probe that reached the telescope [5]. We replay this campaign through OrchScope's orchestration engine at ORION's scale and ask four questions:

1. Does the engine flag a real, centrally commanded campaign, and in which of its phases?
2. How soon does it flag it when run hour by hour, as deployed?

3. How often, under the same settings, does it flag random groups of real scanners on the same port?

4. What does the replay reveal about the engine itself?

Section 2 describes the data, Section 3 the method, Section 4 the results and Section 5 what they show; Section 6 discusses threats to validity and Section 7 reproduction and data use.

2 The Sipscan Data

CAIDA's Sipscan Dataset [5] holds one row per sipscan probe that reached the UCSD Network Telescope, a /8 darknet: 20,255,721 UDP packets to port 5060 from 2,954,108 sources in 208 countries, between January 31 and February 14, 2011. Each row gives the arrival time, the source's country, coordinates and autonomous system, an anonymized source identifier (one to one with the source address), the destination address with its first octet replaced, and the source port. There are no other header fields, and the dataset holds no traffic other than the sipscan's: every source in it is a sipscan bot.

Figure 1 (top) shows the campaign as an ORION-sized telescope sees it: the distinct bots per hour within one /13 of the /8 (Section 3.2), averaged over the 32 such slices. The bots started within minutes at 21:07 on Jan 31 (all times UTC) and scanned for 5.7 days with a daily rhythm. From 13:00 on Feb 6 their number fell within a day to a trickle of 65–238 bots per hour per slice. At 14:00 on Feb 11, 5.0 days after the decline began, the botnet restarted at full strength, peaking at 8,051 bots per hour per slice, and it stopped abruptly at 15:00 on Feb 12, 25 hours later. The near-simultaneous start, restart and stop of a whole botnet are the kind of command that a test of synchrony should see; the steady phase in between is the hard case.

3 Method

3.1 Scope

Without header fields there is no tool fingerprint, so OrchScope's inference engine, which blocks sources by port and fingerprint before clustering them, cannot be evaluated here. We test the orchestration engine alone, on groups whose membership is known: every source of the dataset belongs to the sipscan.

3.2 ORION-sized views

ORION monitors a /13, about half a million addresses. A /8 holds 32 /13s, so we split the telescope into 32 slices by the first five bits of the destination's second octet, which the dataset keeps intact; each slice is an ORION-sized view of the same campaign. Within a slice, a bot is profiled once it has sent two packets there: OrchScope profiles sources with at least two distinct destinations or two packet samples, and a source with two packets has one or the other. For each profiled bot we build the fields that the orchestration engine reads: its hourly activity bitmap over the test window, its number of active hours, its country, and its modal destination, which the residue rule uses. To check this shortcut, we rebuilt 3,000 bots of one slice and window through the production path: ORION-format flow records, one per bot, hour and port, each with up to three packet samples carrying the real destination and source port, fed to the production reducer and profiler. Active hours, countries and ports agreed for all 3,000 bots, and the modal destination for 2,999.

3.3 The test, as in the paper

Every group is tested as in the paper. The statistic is the mean Jaccard co-activity of the members' hourly bitmaps over at most 600 sampled member pairs. The rotation null shifts each member's bitmap circularly

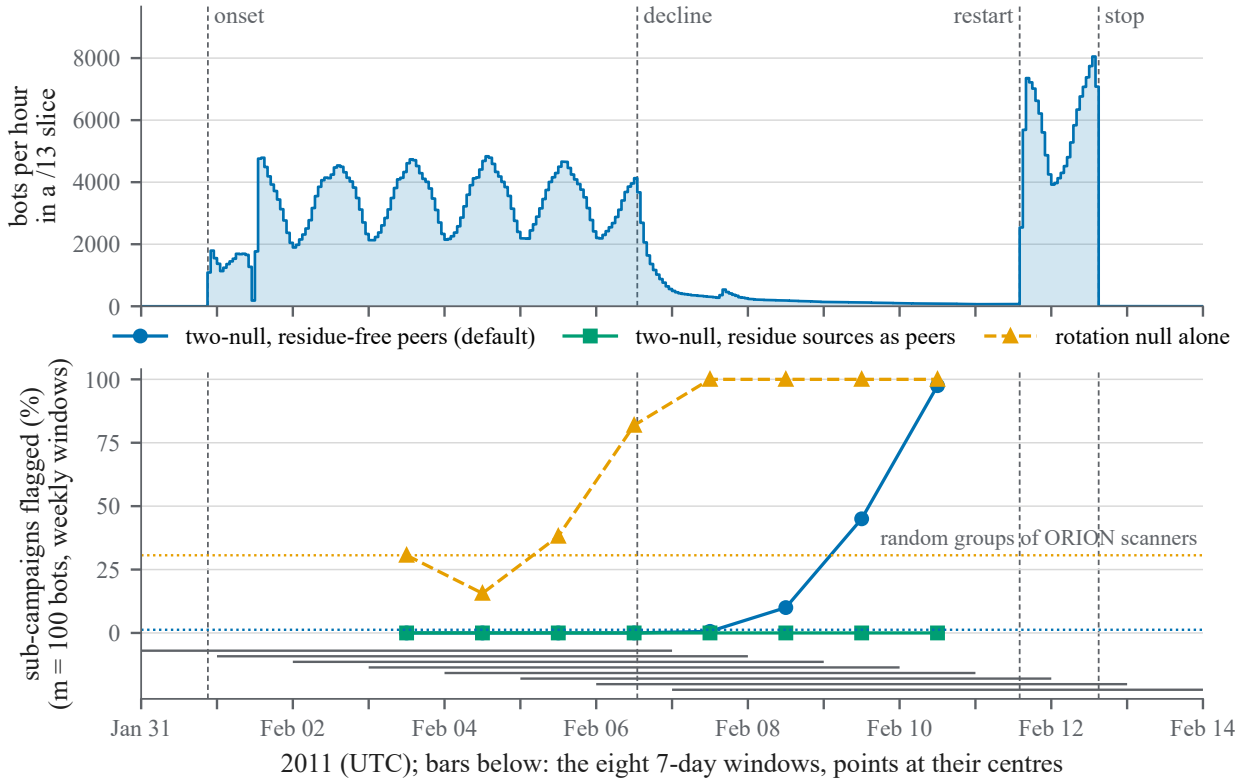


Figure 1: Top: the sipscan as an ORION-sized telescope sees it, as distinct bots per hour in a /13 of the UCSD /8 (mean of the 32 slices); dashed lines mark the onset, the decline, the restart and the stop. Bottom: share of 100-bot sub-campaigns flagged in each 7-day window (bars below the axis; points at their centres) by the two-null test with residue-free peers (the default), by the two-null test with all port-5060 peers, residue sources included, and by the rotation null alone. Dotted lines: random groups of ORION's scanners, two-null test with residue-free peers and rotation null alone.

by an independent whole number of days [6]; the population null replaces each member with a random non-member of the same port and country (of the port alone if the country has fewer than 40) in the same power-of-two bin of active hours. Each null yields an exact Monte Carlo p-value from 200 surrogates, refined with 1,000 fresh surrogates if $p \leq 0.1$ and with 10^4 if $p \leq 0.01$; the two-null p-value is the larger of the two, an intersection–union test [7]. A group whose port has fewer non-members than members is not tested, and its orchestration is undetermined. Weekly windows span 168 hours.

3.4 Peers borrowed from ORION

The population null needs the campaign's port peers, and the dataset has none. We borrow them from the paper's ORION week (September 24–30, 2026): the 21,067 sources whose primary port is 5060. One event dominates that population: 18,793 of its sources (89%) belong to groups that OrchScope sets aside as residue, 18,893 share one modal dark destination, and 18,845 were active in the same single hour. OrchScope never tests residue groups, and it no longer draws peers from their sources either; its first version did, and this replay is why that changed. We therefore run every experiment with two peer sets: the 2,274 sources outside residue groups (*residue-free*, RF), as OrchScope now uses, and all 21,067 sources, residue sources included (RP), as its first version did. Only 6 countries (CN, DE, JP, MY, SG, US) have 40 or more residue-free sources on the port, and they hold 3.8% of the sipscan's bots, so for 96% of the bots the population null draws peers from the whole port. Both datasets keep UTC timestamps, so hours of the day align; in the hourly replay

(Section 3.8), days of the week align too.

3.5 Decision threshold

A group is flagged if Benjamini–Hochberg (BH) at $q = 0.05$ [8] would flag it when its p-value joins the 10,398 p-values that the paper's week tested, that is, if $p \leq 0.0044$. For the rotation null alone, joining the week's rotation p-values, the threshold is $p \leq 0.0257$. Each group thus faces the decision of a deployment in which the sipscan appears among ORION's real campaigns. Each group, sipscan or control, joins the week's family alone, so the share of groups flagged is a per-test rate at the week's threshold, not the false-discovery proportion of one family that holds the sipscan and the controls together.

3.6 Experiment 1: weekly windows

We use 8 weekly windows of 168 hours, starting at 00:00 on each day from January 31 to February 7, in all 32 slices. A slice holds 23,692 to 71,379 profiled bots per window (medians across slices). At that size the sipscan outnumbers its peers in every slice and window, so OrchScope would report it as undetermined; we test it whole with that rule waived, and we test sub-campaigns of $m = 10$, 30, 100, 300 and 1,000 bots drawn at random within a slice and window, 5 per size, slice and window (1,280 per size). For scale, the median orchestrated campaign in the paper's week has 30 sources.

3.7 Experiment 2: negative controls

Random groups of ORION's residue-free port-5060 sources, 320 per size and 1,600 in all, go through the same test, peers and threshold. (Random groups of residue sources are themselves residue, which OrchScope never tests.) We also apply BH to the controls' own p-values, as the paper's negative control does.

3.8 Experiment 3: hourly replay

In deployment, OrchScope updates its campaigns after every hourly batch over a weekly landmark window; the test sees the most recent whole days ending at the current hour, and decisions start once two whole days exist. ORION's week starts on a Thursday, and so do the three windows we replay: from January 27, whose fifth day holds the onset; from February 3, which holds steady scanning and the decline; and from February 10, whose second and third days hold the restart and the stop. At each hourly update from the end of day two to the end of day seven (121 updates per window), the campaign of a slice is every bot profiled in that slice so far; it needs ten members and is testable while its port has at least as many peers as it has members. We also test a random 30-bot sub-campaign of it. The peers are the ORION sources active by the same hour of ORION's week, with their bitmaps cut to the same trailing days.

3.9 Experiment 4: residue peers on the paper's week

To measure what excluding residue sources from the peer pools changes in the paper, we re-test the week's 10,398 campaigns with residue sources admitted as peers, as the first version did, on the pipeline's own per-campaign surrogate seeds, and re-run both negative controls of the paper (same port; same port and country) with those pools.

Table 1: Sipscan sub-campaigns flagged per 7-day window, in percent of 160 per cell: the two-null test with residue-free peers (RF, the default) and with all port-5060 peers, residue sources included (RP), and the rotation null alone (Rot.). "Holds": phases of the scan inside the window.

		$m = 30$			$m = 100$			$m = 1{,}000$		
Window from	Holds	RF	RP	Rot.	RF	RP	Rot.	RF	RP	Rot.
Jan 31	onset, steady, decline	0	0	19	0	0	31	0	0	22
Feb 1	steady, decline, idle	0	0	13	0	0	16	0	0	17
Feb 2	steady, decline, idle	0	0	28	0	0	38	0	0	46
Feb 3	steady, decline, idle	0	0	74	0	0	82	0	0	89
Feb 4	steady, decline, idle	0	0	93	1	0	100	1	0	99
Feb 5	steady, decline, idle, restart	2	0	98	10	0	100	21	0	100
Feb 6	steady, decline, idle, restart, stop	26	0	100	45	0	100	71	0	100
Feb 7	decline, idle, restart, stop	82	0	100	98	0	100	100	0	100

4 Results

4.1 Weekly windows

Table 1 and Fig. 1 (bottom) give the flagged share per window. With all port-5060 sources as peers, residue sources included, no sipscan sub-campaign is flagged in any window or at any size (0%): against peers this synchronized, the population null cannot reject. With residue-free peers, the two-null test flags the sipscan in the windows where its bots switched on and off: 82% of 30-bot and 98% of 100-bot sub-campaigns in the window from February 7, which holds the restart and the stop but no steady scanning, and 26–71% in the window from February 6. In the four windows that the steady phase dominates, from January 31 to February 3, it flags none (0%). A weekly window also dilutes the onset, which the window from January 31 holds together with 5.7 days of steady scanning; the hourly replay isolates it (Section 4.3). The rotation null alone flags 13–100% of the sipscan's sub-campaigns, depending on the window.

Taken whole, the sipscan is undetermined in all 256 slice-windows (0 testable), as it outnumbers the port's peers. With that rule waived, the population null rejects in 0% of the slices in the window from January 31 and in 53% in the window from February 7; the rotation null alone rejects in 9% and 100%, and in every slice from the window of Feb 4 on.

4.2 Negative controls

Table 2 pools the windows by size. Random groups of ORION's port-5060 scanners are flagged 5 times in 1,600 (0.3%) with residue-free peers and 20 times (1.2%) with residue sources as peers. With BH applied to the controls alone, as in the paper, none is flagged with either peer set. The rotation null alone flags up to 35% of them at the deployment threshold, and 212 of 1,600 under their own BH: alone, it cannot tell the sipscan from the port's independent scanners. The controls' members are mostly always-on scanners (median 168 active hours of 168, against 2 for the sipscan's bots), as is the port's residue-free population.

4.3 Hourly replay

Figure 2 shows the share of slices whose 30-bot sub-campaign is flagged at each hourly update. With residue-free peers the onset is caught at once. At the first update after it, at 22:00 on Jan 31, 14 slices already hold a campaign of ten or more bots and 14 of them are flagged; from 23:00 on Jan 31 on, all 32 are. The whole campaign stays testable until 14:00 on Feb 1, when it outgrows its peers and becomes undetermined, while its 30-bot sub-campaigns stay flagged in 91% of the slice-updates to the window's end. In the window

Table 2: Flagged share in percent, over all eight windows, by group size: sipscan sub-campaigns ("sipscan") and random groups of ORION's residue-free port-5060 sources, under the same test, peers and threshold (RF: residue-free peers, the default; RP: all port-5060 peers, residue sources included; Rot.: rotation null alone).

	sipscan			random groups		
m	RF	RP	Rot.	RF	RP	Rot.
10	1.4	0.0	30.3	0.0	0.0	2.2
30	13.7	0.0	65.6	0.3	0.9	18.8
100	19.1	0.0	70.8	1.2	2.2	30.6
300	22.3	0.0	70.3	0.0	1.2	31.6
1,000	24.1	0.0	71.6	0.0	1.9	35.3
groups per row	1,280			320		

Table 3: The paper's week (residue-free peers, the default) and the same week re-tested with residue sources admitted as peers, on the pipeline's own surrogate draws, with the paper's two negative controls run with each pool.

	Residue-free (default)	Residue sources
Testable campaigns	8,363	8,781
Orchestrated (BH, $q = 0.05$)	906	884
flagged by both	864	864
TCP / UDP	477 / 341	465 / 330
Same-port control: flagged / groups	0 / 8,000	0 / 8,059
uncorrected $p < 0.05$ / $p < 0.01$	2.7% / 0.5%	2.8% / 0.5%
Same-country control: flagged / groups	0 / 7,443	0 / 7,371
uncorrected $p < 0.05$ / $p < 0.01$	2.8% / 0.6%	2.8% / 0.8%

from February 3, 2 of 3,872 slice-updates flags a sub-campaign, whereas the rotation null alone flags up to 81% of them once the decline enters the window. The restart falls on day two of its window, so the first decision comes at 00:00 on Feb 12, 10 hours after it; that decision flags the 30-bot sub-campaigns in 32 of 32 slices, while the whole campaign is by then undetermined.

With residue sources as peers, the onset is flagged in at most 18 of 32 slices (at 00:00 on Feb 1). The restart is flagged in 97% of slice-updates until 11:00 on Feb 12, the hour of ORION's week at which its residue event enters the peer pool, and never after it. That the restart's first decisions precede the event is a coincidence of aligning 2011 with ORION's week.

4.4 Residue peers on the paper's week

Table 3 compares the paper's week with the same week re-tested with residue sources admitted as peers. Residue-free pools leave fewer campaigns testable (8,363 instead of 8,781), since some ports lose peers, and 906 campaigns orchestrated instead of 884: 864 are flagged either way, 42 only with residue-free peers and 20 only with residue peers. Both negative controls flag none either way, with practically unchanged uncorrected rejection rates.

5 What the Sipscan Shows

The engine detects commanded synchrony. The two-null test flags a real C&C-orchestrated campaign where the C&C switched its bots on or off together: at the first or second hourly update after the sipscan's

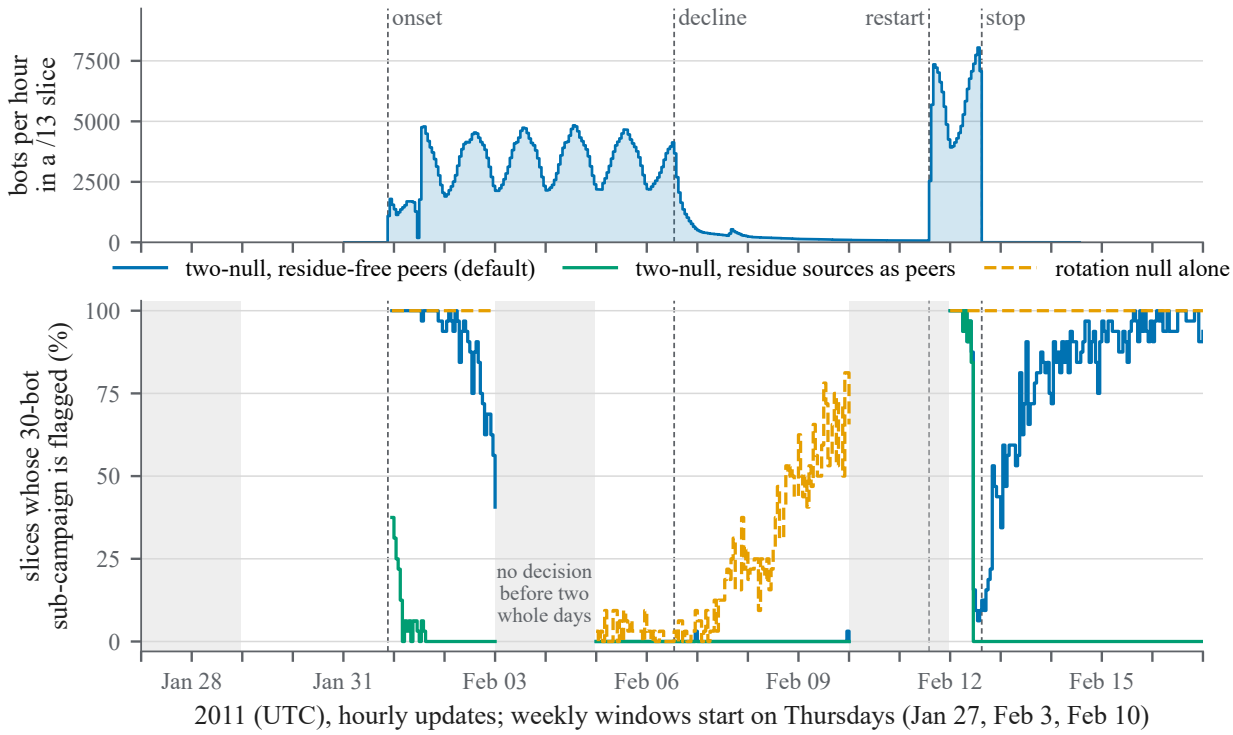


Figure 2: Hourly replay in the deployment mode. Top: as in Fig. 1. Bottom: share of the 32 slices whose 30-bot sub-campaign is flagged at each hourly update of the three Thursday windows; shaded: the first two days of each window, when no decision is made.

onset at ORION's scale, and at the first decision after its restart. This is the synchrony the test is built to find, and the replay shows it on a real botnet rather than a simulated one.

A steady phase is invisible to a timing test. During its 5.7 days of steady scanning, each bot appears in an ORION-sized view for about two hours, and the botnet's volume follows the same daily rhythm every day. Its timing then matches that of independent scanners, and no test of synchrony can separate the two; whatever coordinated the bots in this phase did not show in when they ran. For such campaigns OrchScope reports the campaign without the orchestration flag, so "not orchestrated" means "no evidence of synchrony", not "independent".

The rotation null alone is not enough. It flags much of the sipscan, but also up to 35% of random groups of the port's scanners, as the paper's negative controls found across all ports.

Peer pools must exclude residue sources. The engine's first version included them. On a port that one residue event dominates, it then compared every campaign with peers more synchronized than any scanner, and here it hid every sipscan sub-campaign. OrchScope now excludes them, which keeps both of the paper's negative controls clean and changes its week from 884 to 906 orchestrated campaigns (Table 3).

6 Threats to Validity

Transplanted peers. The peers come from ORION in 2026 and the sipscan from UCSD in 2011; their port-5060 populations differ, and the residue event of ORION's week is specific to that week. The pairing

strains only the population null, whose exchangeability assumes peers active in the same weeks as the bots; the rotation null uses each bot's own activity. Other peer populations could change the results.

Sub-campaigns. The tested groups are random subsets of one botnet, not groups formed by OrchScope's blocking and clustering, which the dataset cannot exercise; taken whole, the campaign is untestable at ORION's scale.

Dependence between slices. The 32 slices are views of the same bots, not independent replicates; the flagged shares describe the spread across views, not independent trials.

One campaign, one scale. All results hold for one campaign and one telescope size (/13). Weekly windows align hours of the day between 2011 and 2026, and the replay also aligns days of the week; the restart result with residue peers depends on that alignment.

7 Reproduction and Data Use

All code and aggregate results are in the OrchScope repository, `https://github.com/ebh2024ebh/orchscope`:

- `code/sipscan_eval.py`: the analysis, with subcommands `prep`, `check`, `week` and `stream`, run next to the paper's week profiles (Experiments 1–3);
- `code/revision_checks.py respeers`: Experiment 4 (`PEER_NORES=0` runs the whole pipeline with residue peers);
- `code/sipscan_timeline.py`: the hourly counts of Fig. 1;
- `code/sipscan_doc.py`: this report's figures, tables and numbers, from the aggregate results in `results/`; no number is typed by hand.

All random draws use fixed seeds, so reruns reproduce every number.

The sipscan data are used under CAIDA's Acceptable Use Agreement for publicly accessible datasets and are not redistributed; the released files hold aggregates only. ORION traffic was analyzed passively under a data-use agreement with Merit Network and is not redistributed either.

References

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